

Decoding Two-Class Motor Imagery EEG with Capsule Networks

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Abstract—Recently, deep learning approaches such as convolutional neural networks (CNN) have been widely applied to improve the classification performance of motor imagery-based brain-computer interfaces (BCI). However, CNN is known to have a limitation that its classification performance is degraded when the target data are distorted. Particularly in case of electroencephalography (EEG), the signals measured from the same user are not consistent. To address this issue, we propose to apply Capsule networks (CapsNet) which implicitly learn various features, thereby achieving more robust and reliable performance than traditional CNN approaches. In this paper, a novel method based on CapsNet to classify two-class motor imagery signals is presented. The motor imagery EEG signals are transformed into time-frequency images using Short-Time Fourier Transform (STFT) and then supplied for training and testing capsule networks. The experimental results on BCI competition IV 2b dataset show that the proposed CapsNet based architecture outperforms previous CNN-based approaches.

Keywords—capsule network, brain computer interface, EEG, deep learning, motor imagery classification

I. INTRODUCTION

Currently, various approaches such as a gesture-based control and an eye tracking method have been proposed to improve the user experience while interacting with devices [1], [2]. In particular, the Brain Computer Interface (BCI) technology has been actively studied for people with disabilities to use the devices more efficiently and comfortably.

The BCI is a promising technology that can help users directly control the machines without any physical interaction by converting the electroencephalogram generated from the neuron's neurotransmission activity into a control instruction through a sequence of preprocessing, feature extraction and translation processes. The most commonly used signals in BCI are Event-Related Potential (ERD), Steady State Evoked Potential (SSEP), and Sensorimotor rhythm (SMR). Among the EEG signals, SMR which occurs when there are motor imagery and movement (e.g., right-hand, left-hand, foot, and tongue) is a popular signal and commonly used in many BCI applications. In the SMR-based BCI, Common Spatial Pattern (CSP) [3], Sub-Band Common Spatial Pattern (SBCSP) [4],

and Filter Bank Common Spatial Pattern (FBCSP) [5] techniques were used to classify the motor imagery EEG.

The CSP is a spatial filter algorithm that maximizes the variance between the classes and contributed greatly to the performance improvement of a motor imagery EEG classification. However, the CSP suffers from the limitation that the frequency band must be manually selected for individual characteristic. To solve this problem, SBCSP and FBCSP have been proposed. The SBCSP and the FBCSP algorithms solved the problem of CSP by setting several fixed frequencies of the filters. However, a classification performance may be degraded if appropriate filters including active frequency bands are not selected.

In order to improve the classification performance, deep learning approaches such as CNN have been developed. The study from [6] presented a way to optimize CNN network based on motor imagery EEG. This approach compared and analyzed the accuracy according to the network structure and types of non-linearity. However, the CNN also has its own limitations. First, a classification performance degrades when the target data are distorted (e.g., tilting or rotating). Second, the CNN uses a max pooling to reduce the spatial size of feature maps to broaden the field of view so that it can take spatial information to some extent, but does not consider the core spatial layer between simple objects and complex objects. To overcome these limitations, Hinton et al. presented a new type of deep-learning architecture called CapsNet [7]. The CapsNet tries to solve this problem by learning various features with the theory of a capsule (e.g., tilting, rotating, width for image domain) and taking into account the core spatial layer between simple objects and complex objects.

The data in the EEG domain is not consistent even if measured from the same person for the same task on the same day. In this paper, we propose a novel method for applying CapsNet to classify two-class motor imagery EEG signals. We exploit the CapsNet to learn various latent features from the inconsistent signals for the motor imagery task and then demonstrate if the CapsNet-based architecture can successfully decode the signals. To this end, the raw signals of motor image EEG are converted to 2D image by the STFT algorithm which transforms a raw signal into a time-frequency domain. The

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converted 2D images are then supplied for training and testing the CapsNet architecture.

II. CAPSULE NETWORK

A Capsule is a group of neurons whose activity vectors represent various properties of an entity and the length of the vector indicates the probability that the entity exists. In the CapsNet, a max-pooling algorithm is replaced with a dynamic routing by agreement algorithm to achieve a robust and reliable performance.

Each capsule can find a prediction for a parent capsule, and if the prediction matches the actual output of the parent capsule, the coupling coefficient increases. If u_i is the output of capsule i , the prediction for parent capsule j is calculated as:

$$\hat{u}_{j|i} = W_{ij} u_i$$

where W_{ij} is a weight matrix. The coupling coefficient c_{ij} are calculated using the softmax function:

$$c_{ij} = \frac{\exp(b_{ij})}{\sum_k \exp(b_{ik})}$$

where b_{ij} is a log prior probability that capsule i should be coupled to parent capsule j , and its initial value is set to 0. The input vector of the parent capsule j is therefore calculated as:

$$s_j = \sum_i c_{ij} \hat{u}_{j|i}$$

Finally, the output v_j is calculated using a squash function which acts as a non-linear activation function like ReLU in CNN:

$$v_j = \frac{\|s_j\|^2}{1 + \|s_j\|^2} \frac{s_j}{\|s_j\|}$$

With this squash function, the output of a capsule is shrunk to 0 when the capsule's output vector is small and to 1 when the capsule's output vector is large. Finally, the entity's probability of occurrence can be expressed as a value between 0 and 1. The log probabilities b_{ij} is updated through n routing iterations by adding the following agreement:

$$a_{ij} = v_j \cdot \hat{u}_{j|i}$$

Finally, the loss function of the CapsNet is defined as:

$$L_k = T_k \max(0, m^+ - \|v_k\|)^2 + \lambda(1 - T_k) \max(0, \|v_k\| - m^-)^2$$

where L_k is loss of each class k . If the input data matches the label, the T_k value is 1 and 0 otherwise. The hyper parameters m^+ and m^- are set to 0.9 and 0.1, respectively. λ serves to reduce the effect of loss on labels that do not belong to the correct class and has a value of 0.5.

As a regularization method, a reconstruction process through 3 fully connected layers with the output vector is performed. The reconstruction loss between the original image and reconstructed one is computed and generally reduced by 0.0005 in order not to dominate a margin loss during training.

III. THE PROPOSED METHOD

In this section, we describe an overall architecture of proposed system and the details for preprocessing EEG signals, construction of capsule networks and training procedure.

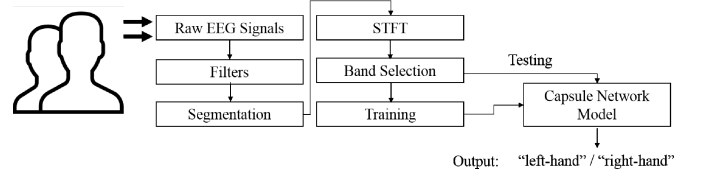


Fig. 1. Overall workflow of the proposed system

The overall workflow of the proposed method is depicted in Fig. 1. First, EEG signals are measured from a set of electrodes (i.e., channels) attached to the user scalp. The raw EEG signals are then notch filtered at 50Hz for noise reduction purpose. After noise reduction, segmentation is applied to signals to determine the length of the data (e.g., 2 seconds since visual cue) for further processing.

As deep learning-based approaches such as CNN and CapsNet generally works well for image classification tasks, we transform the 1D EEG signals into 2D images using STFT algorithm. With the STFT algorithm, we can generate one 2D time-frequency domain image from each channel (i.e., electrode). For example, if we have EEG signals from 3 electrodes, then 3 channel 2D image data (like RGB image) can be generated and used for training and testing classifiers. More details about the generation of 2D STFT images measured from 3 electrodes with sampling rate of 250Hz are described in Fig. 2. After EEG signals get notch-filtered and segmented according to the procedure described above, we get data array of size 3 x 1 x 500. Applying the STFT algorithm, we have three 2D data array of size 76 x 14, where Y-axis and X-axis indicate the frequency and time, respectively.

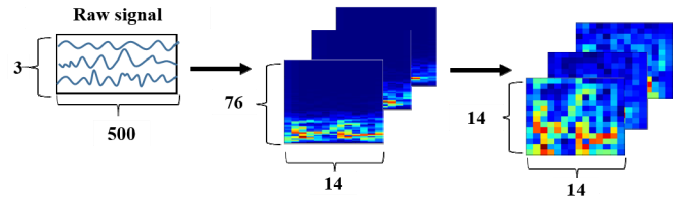


Fig. 2. Generation of 2D input images using STFT

The EEG signals are divided into bands by frequency as follows: alpha (8-15Hz), beta (16-31Hz), gamma (>32Hz), theta (4-7Hz), delta (<4Hz), and mu (8-12Hz). Depending on the purpose and application of the study, only a specific range of EEG signals (e.g., mu and beta bands) are selectively used. For motor imagery classification tasks, frequency bands of 5-30Hz are commonly used. In this study, we chose the frequency range between 8-30Hz to extract STFT spectrogram images. Finally, we have 3 channel 14x14 2D images to be fed to capsule networks for training and testing.

Fig. 3. illustrates the network structure of the proposed CapsNet-based approach. Similar to the original CapsNet paper [7], our CapsNet also takes 2D images as initial input. In our architecture, there are two convolution layers (kernel size: 6x6

with 1 stride 1) and one dropout layer before the PrimaryCaps layer, which results in 32 primary capsules each of which consists of 8 dimensions. The last layer called MI (Motor Imagery) Caps has 16 dimensions per class (i.e., left-hand vs right-hand). The training of CapsNet is done by dynamic routing by agreement between the capsules as explained above. The learnability and feasibility of the CapsNet on EEG domain are discussed in Section IV.

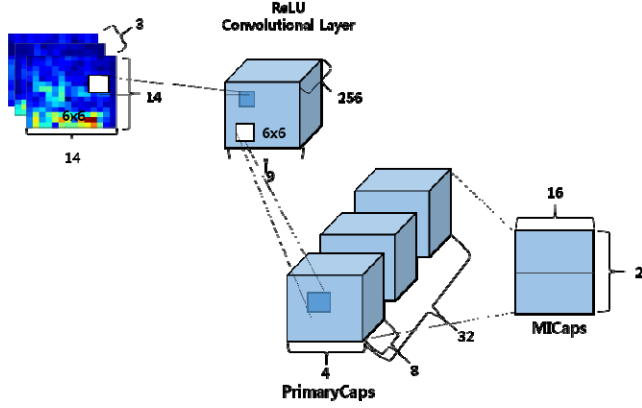


Fig. 3. Network structure of the proposed method

IV. EXPERIMENTS

A. Dataset and Preprocessing

To evaluate the performance of the proposed CapsNet-based approach on BCI dataset, we have conducted an extensive experiment on BCI competition IV 2b dataset [8]. The dataset consists of EEG signals measured from 9 subjects during two-class motor imagery tasks (i.e., left-hand and right-hand). Three bipolar channels (C3, C4, and Cz) are used for EEG recording with sampling frequency of 250Hz. Also, the signals were bandpass filtered between 0.5Hz and 100Hz and a notch filter at 50Hz was applied. The dataset consists of five sessions where the first three sessions are for training and the rest are for testing. On average, training data for each subject has 400 trials (120+120+160) whereas test data has 320 trials (160+160). According to the experiment guideline of the dataset [8], a classifier is trained and tested for each subject. As motor-imagery activates sensorimotor systems that can be captured in μ and β bands, a broad frequency band (i.e., 8Hz ~ 80Hz) are typically chosen for feature extraction [3]. It is also well known that EEG segments from 0.5s to 2.5s after cue onset generates better classification results. Therefore, in our experiment the original EEG signals from the dataset have been bandpass filtered between 8~30 Hz and only two seconds of EEG segments (0.5~2.5s after cue on-set) are used.

The networks used for our experiments are implemented using BrainDecode framework [6] which provides various features for EEG processing. The experiments are conducted with a desktop PC equipped with Nvidia Pascal Titan X GPU.

B. Performance Evaluation

First, we evaluated the feasibility of applying the capsule network to EEG motor imagery classification, which is

originally designed for image and object classification. The capsule network for each subject is trained with 32 primary capsules whose dimension is 8 and 16-dimension MI Caps with 2 outputs (i.e., left vs right).

Fig. 4 shows if the capsule network can be well trained on EEG domain with STFT images. As can be seen, both the train and test loss decreases rapidly within 10 epochs and gradually converges to 0. The performance in terms of classification accuracy of the proposed method is summarized in Table I. In this study, the classification accuracy indicates 1 – misclassification rate (i.e., the number of misclassified trials out of the entire trials). The capsule network achieves 77% of classification accuracy when the number of routing iteration is set to 3. It was also found that there does not exist a significant effect of the number of iterations on the classification accuracy of the network. We also investigated the effect of a reconstruction process on the performance. As shown in Table II, there is a slight performance improvement (0.24% point) over the one without reconstruction.

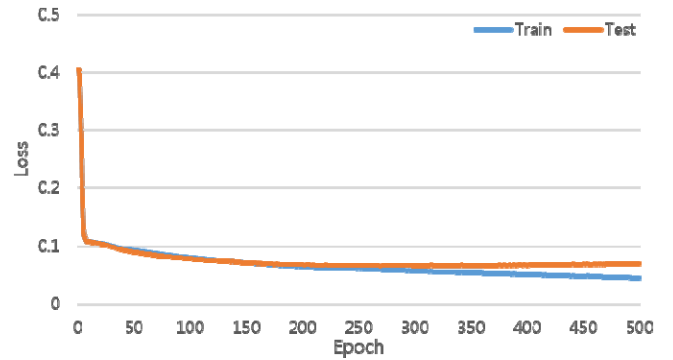


Fig. 4. The change of loss during training and testing

TABLE I. BRIEF SUMMARY OF CLASSIFICATION ACCURACY

Routing – 1	Routing – 2	Routing – 3
76.96%	76.91%	77.00%

TABLE II. EFFECT OF RECONSTRUCTION ON CLASSIFICATION ACCURACY WITH ROUTING ITERATION OF 3

With Reconstruction	Without Reconstruction
77.00%	76.76%

In addition to the investigation on the learnability of capsule networks for EEG domain, we compared the classification accuracy of the proposed capsule network-based approach with a traditional CNN-based method. For comparison, we implemented a CNN network suggested by [9] which has 2 convolution and pooling layers for decoding two-class motor imagery signals with STFT images. As an input for training and testing the CNN method, the same STFT spectrogram images with 3 channel 14x14 2D vectors are used. In this experiment, the classification accuracy was calculated by averaging accuracy scores for each subject obtained from 10 repetitive evaluations.

TABLE III. COMPARISON OF CLASSIFICATION ACCURACY

Subject	CNN	CapsNet (The proposed)
#1	69.78%	74.09%
#2	54.75%	53.64%
#3	52.88%	51.72%
#4	95.31%	96.00%
#5	85.91%	86.53%
#6	78.03%	82.59%
#7	69.75%	72.22%
#8	87.56%	89.31%
#9	80.91%	86.91%
Average	74.99%	77.00%

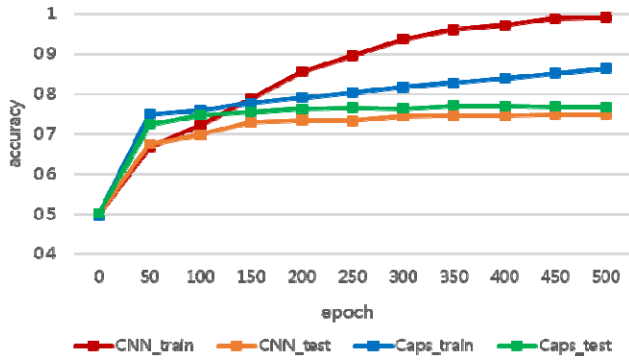


Fig. 5. Classification accuracy on training and test dataset according to the epochs

The summary on the comparison of classification accuracy between the proposed CapsNet-based method and CNN-based method is presented in Table III. We report the classification accuracy of each subject as well as the average classification accuracy of all subjects. As can be seen from Table III, the proposed CapsNet-based method outperforms the CNN-based method with average 3.01% of performance improvement. Specifically, the CapsNet-based method works better than the CNN-based approach except subject No.2 and No. 3. The evaluation on subject No.4 achieved the best result (96.00%), whereas subject No.3 produced the worst result (51.72%). On the other hand, the highest performance improvement (6.00%) was achieved from subject No. 9.

Fig. 5 depicts the changes of average classification accuracy on training and test dataset according to the epochs. From this figure, we can find some meaningful insights:

1) The classification accuracy of CNN-based method on training dataset converges to almost 1.00 at the 500th epoch, which is significantly better than CapsNet-based method (0.88). However, it is worth noting that the classification accuracy on test dataset of CapsNet-based method is higher than that of CNN-based method (i.e., 77.0% vs 74.99%). This implies that the CNN-based method is overfitted to the training dataset; therefore, presented worse results on test dataset. This can be caused from the nature of the CapsNet which implicitly tries to

learn various features so as to produce robust and reliable performance.

2) The proposed CapsNet-based method more quickly converges than the CNN-based method. As depicted in Fig. 5., the classification accuracy of the CapsNet-based method sharply increases until the 50th epoch and then gradually increases for training and gets stable for testing. However, the CNN-based method takes more time to have a stable performance.

The experimental results on the learnability of the CapsNet-based method and the comparison with the CNN-based method show the CapsNet is promising for EEG domain and requires much more investigation on network structure and parameter optimization for performance improvement.

V. CONCLUSIONS

In this paper, we proposed a method to apply Capsule Network for decoding two-class motor imagery EEG signals. To evaluate the performance and feasibility of the proposed method, several experiments on BCI competition IV 2b was conducted. The experimental results show that the proposed method outperforms previous CNN-based approaches on two-class motor imagery tasks. Furthermore, there is a still room for improvement of CapsNet-based approach for EEG decoding. In the future, we will expand our network architecture to handle multi-class motor imagery classification tasks.

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